

An Intelligent Retrieval Augmented Generation Chatbot for Contextually-Aware Conversations to Guide High School Students

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Abstract— The growth of Large Language Models (LLM) has showcased both their potential and limitations in domain-specific applications. A significant limitation is their tendency to confidently present false information, known as hallucination, which compromises the integrity of their outputs. Retrieval Augmented Generation (RAG) systems mitigate this by enhancing the Large Language Models' ability to retrieve accurate, source-specific knowledge. Many American students begin preparing for college applications in middle school, seeking guidance on planning their high school years to achieve personal goals. The diversity in students' interests, skills, and circumstances often results in an overwhelming amount of information and advice. This research introduces a novel RAG chatbot designed to provide personalized assistance to high school students using their course guide as the primary source document. By integrating a high school course guide with a RAG system, the chatbot delivers relevant suggestions, helping students navigate their educational paths with greater confidence and precision. This system aims to minimize the risk of hallucination by ensuring that responses are grounded in the accurate and relevant information contained within the course guide. Consequently, it addresses the crucial need for personalized educational planning, enabling students to make informed decisions that align with their unique aspirations and circumstances. This research highlights the transformative potential of RAG systems in educational contexts, offering a reliable tool to support students in their academic and personal growth.

Keywords— Retrieval Augmented Generation (RAG), Context Retrieval Pipeline, Ground Truths, Hallucination, Vectorstore, Semantic Similarity Search, Faithfulness, Context Precision, Context Recall, Answer Relevancy

I. INTRODUCTION

The admission rates for highly rated US colleges are at an all-time low, making acceptance into top colleges increasingly tricky. These colleges use an admission review portal for evaluating students' academic success and standardize test scores, extracurricular activities, personal statements, and letters of recommendation. Consequently, US students start planning their college journey from middle school, participating in activities competitions, and planning their high school courses. Finding relevant information, such as which courses impress colleges for specific majors and which extracurriculars stand out, is crucial for improving students' chances of acceptance. Students' backgrounds, such as being a first-generation applicant, an immigrant, or even

the first child in the family, significantly affect the students' access to reliable and accurate information which further complicates their ability to create a plan for high school. Students struggle to find relevant information to help them plan their high school journey because there is a tremendous amount of information on how to prepare a successful college application, to the point it is difficult to find information relevant and accurate for each student. This is especially problematic if they are from backgrounds unfamiliar with the constantly changing college applications. Students from these backgrounds also especially struggle in finding relevant information because they don't know what to look for due to a lack of context.

Large Language Model models [1] effectively solve these challenges because their ability to understand and respond in natural language allows them to be content-aware, synthesize disconnected information, and thereby adapt to produce highly personalized experiences. Large Language Model model's strength in content awareness [2] enables them to analyze the context of a student's query, understand questions, and provide personalized responses. For instance, when a student asks about the best high school courses for a particular college major, a Large Language Model can offer detailed and specific advice to that student's background and goals. This includes the recommended courses and additional relevant information, such as prerequisite knowledge and potential career paths. Additionally, synthesis outputs comprehensive answers that draw on vast knowledge, which is particularly beneficial for students who need guidance on multiple aspects of their high school and college preparation. Finally, to touch on Large Language Models' adaptability, using domain-specific sources such as a high school course guide can enhance its ability to provide precise and actionable advice based on a student's school's specific offerings and requirements.

However, Large Language Models can sometimes present false information confidently, known as hallucination [3], which can mislead students seeking reliable guidance. Additionally, Large Language Models struggle with long-term information retention, meaning they might not consistently recall previous interactions or specific details over time, disrupting the continuity and accuracy of the advice provided. RAG systems [4] address these issues using vector embeddings and databases to simulate long-term memory, enabling accurate recall from predefined knowledge, which mitigates the risk of hallucination.

Furthermore, by maintaining context over long-term interactions, the system can ensure continuity and coherence in information provided.

Since large language models can understand natural language and RAG can maintain contextually and factually relevant information [5], this integrated approach facilitates a clearer and more comprehensive understanding of the high school planning process, ultimately helping students make informed decisions based on precise and contextualized information.

The paper's organization is as follows: Section II presents a broad survey of related works. Section III elaborates on the proposed model's workings. In section IV, the paper records the experimental results and discusses possible case studies with respect to the considered problem statement. Finally, section V concludes the paper and proposes future directions.

II. RELATED WORKS

AI-Based Generative Pre-trained Transformers (GPT), including ChatGPT of OpenAI and Bard of Google, are becoming popular in many industry sectors, including Education, Research, and Publications. GPT models can achieve superior performance in natural language interpretation and generating tasks by utilizing extensive pre-training on various datasets. Aithal et al. [6] discussed the capabilities of GPTs in various research-related tasks such as content summarization, literature review assistance, language translation, data analysis, peer review assistance, identification of research gaps, tailoring content for different audiences, continuous learning, and updates. The study also highlights the limitations of GPTs, including potential summarization errors, translation inaccuracies, and the need for human oversight to ensure quality and accuracy.

Neyem et al. [7] performed a study with 150 students. They considered the strategies such as direct searches in the lessons learned database, queries to a generative pre trained transformers (GPT) model, and query enrichment with lessons. query enhancement with Stack Overflow data before GPT processing. In addition, the analysis demonstrated that texts generated by the Large Language Model closely mirrored the linguistic standards and topical relevance of university course requirements. However, their model has the limitation of delivering generic recommendations not done to the user's profile.

William and Altamimi [8] suggest a unique method for implementing text embedding that combines a large language model with the Retrieval Augmented Generation (RAG) model. Conventional techniques for text embedding frequently depend on pre-trained sentence encoders or word embedding, which are not always effective in capturing the text's semantic meaning. The suggested method uses the RAG model, which blends generation-based and retrieval-based methods for text embedding, to get beyond these restrictions. The study offers a broad framework for improving the semantic relevance of context chunks; however, they have not optimized the model to run with domain specific knowledge.

Li et al. [9] proposed an end-to-end system design that utilizes Retrieval Augmented Generation (RAG) to improve the factual accuracy of large language models (LLMs) for domain-specific and time-sensitive queries related to private knowledge bases. They used this RAG approach with test

web data in CMU websites collected using web crawlers. Their experiments demonstrate the system's effectiveness in generating more accurate answers to domain-specific and time-sensitive inquiries. The study also revealed the limitations of fine-tuning LLMs with small-scale and skewed datasets.

The study conducted by Pokhrel and Banjade [10] represents the development of a content generation tool based on the open AI language model by utilizing GPT 3 in the backend as an API to generate the necessary information for the model. This content generation tool uses a recurrent neural network (RNN) architecture, which enables it to make more accurate predictions than rule-based chatbots. However, RNNs have been overtaken by a new neural architecture known as transformers. It surpasses its predecessors in performance, scalability, and state-of-the-art results in handling complex language understanding and generation tasks.

Udayan et al. [11] developed a Chatbot using the Long Short-Term Memory (LSTM) architecture to respond to student questions. This chatbot was trained on keywords used by the bot to get to the context of the question asked by the user. Here, when the user enters a query, it captures the keywords present in the text, and particulates the question being asked, which can find and output the answers defined by a user. However, LSTM systems have been extended and updated by transformer architecture because of the ability of transformers to better perform natural language processing tasks key to providing an adaptable and effective to the end users. Additionally, their chatbot does not use LLM for generation purposes, significantly decreasing their system's ability to understand the question and grasp the needed context to retrieve the relevant information and provide better answers.

Miladi et al. [12] investigated using RAG to improve the accuracy of GPT models in educational settings, particularly Massive Open Online Courses (MOOCs). A comparative analysis of various GPT model iterations showed a significant improvement in accuracy using the RAG-augmented GPT-4. This enhancement highlights the considerable potential of RAG-augmented GPT models in improving the accuracy of content generation.

Maryamah et al. [13] created a Retrieval-Augmented Generation-Based chatbot using the OpenAI Ada embedding model and the GPT-3.5 - Turbo model for response generation. They investigated the difference between the similarity search and MMR approaches for retrieving context from a vector store and found no significant difference in chatbot quality. This experiment aimed to optimize and not evaluate the chatbot's effectiveness in a specific domain or use case.

Thway et al. [14] developed Professor Leodar - a custom-built, Singlish-speaking Retrieval Augmented Generation (RAG) chatbot designed to integrate Large Language Model capabilities to improve education. Deployed at Nanyang Technological University, Singapore, Professor Leodar offers contextually relevant information which helps students with content learning, engagement, and exam preparedness. Professor Leodar's deployment also highlighted the benefits of highly personalized and contextually aware responses in an educational setting. Their system uses Claude 3 and primarily responds with Singlish; however, since English is a

subset of English that most LLMs are not specifically trained on, a use case focused on base English will likely yield better generative results.

Odede and Frommholz [15] presented JayBot, an LLM-based chatbot system to enhance the user experience of prospective and current students, faculty, and staff at a UK university. The objective of JayBot is to provide information to users on general inquiries regarding course modules, duration, fees, entry requirements, lecturers, internship, career paths, course employability, and other related aspects. Leveraging the use cases of generative artificial intelligence (AI), the chatbot application was built using OpenAI's advanced large language model (GPT-3.5 turbo); to tackle issues such as hallucination as well as focus and timeliness of results, an embedding transformer model has been combined with a vector database and vector search. This demo showcases the effectiveness of LLM models combined with RAG systems in retrieving relevant information to aid students in educational contexts that are often difficult to navigate due to the novelty and large amount of information.

Identified Research Issues

One key issue identified is the challenge new high school students face in accessing relevant information to make informed decisions and set long-term goals, often due to a lack of contextual understanding. Traditional keyword-based search methods fall short because they rely solely on matching keywords, often leading to incomplete or irrelevant results. In contrast, a Large Language Model (LLM) interprets user queries more holistically, understanding the intent behind the question rather than just focusing on specific terms. This allows the LLM to provide relevant background information, clarify ambiguities, and deliver more precise, context-aware responses, ensuring students receive actionable guidance. This research focuses on building an RAG chatbot to address this gap by leveraging the LLM's capabilities [16, 17]. We will evaluate the quality of the chatbot's responses to ensure they meet the necessary standards before conducting tests with actual students.

III. PROPOSED WORK

Figure 1 represents the proposed architecture of the RAG Recommender Pipeline. This pipeline consists of two main segments: retrieval and generation. The retrieval segment consists of source document preprocessing and similarity search. The generation segment comprises prompt preparation and the Large Language Model for response generation.

Retrieval

Preprocessing

The preprocessing step prepares source documents, such as the high school course guide PDF, to be programmatically searched for context retrieval. After the PDF has been loaded into the program, the text content is read and extracted into one string stored in short-term runtime memory. This large string is then split into chunks of a set size. A chunk overlap is also defined to ensure no text is skipped over. This splitting process is essential because when the most relevant chunks are retrieved later, they will be passed as part of the prompt to the Large Language Model, so if they are too large, then the final prompt during the generation stage would have too many tokens for the Large Language Model to process. Each chunk is passed through a vector embedding

algorithm to convert the plaintext into vector representations. These vector-embedded chunks are then stored in a vector store database. This step must occur before the user asks a question because it sets the source document as the context.

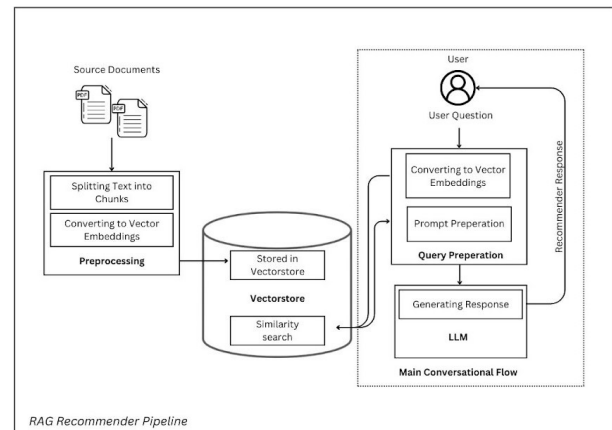


Fig. 1: Proposed architecture for RAG Recommender Pipeline

Query Preparation

The query preparation step uses the user question to retrieve the relevant context chunks which are then synthesized into a prompt passed to the Large Language Model during the generation segment. The user can enter a question through the front-end user interface. This plaintext user question is run through the same vector embedding algorithm as the source document chunks to convert the text into vector embeddings. This vector embedding is sent to the vector store for semantic similarity search. It retrieves the most relevant chunks and sends it back to the program to be sent to the generation segment.

Generation

Prompt Preparation

The user question and retrieved chunks are converted to plain text and then concatenated to a predefined prompt template (Figure 2). The prompt template uses rule-based prompt engineering to ensure the Large Language Model's responses are in the format and consistent with what the student (i.e., the end-user) expects.

```

# prompt engineering
template = """
You are a helpful, respectful and honest assistant.
Always answer with just enough information as necessary, and
remember to format your answers, such as with bullet points, if needed.
Ask any questions to the user if you need clarifying information
to satisfactorily answer their question.
If you don't know the answer or the information is not provided in the
source documents, say that you don't know. Do not share false information.

Question: {question}
Context: {context}
Answer
"""
    
```

Fig. 2: Prompt Template

Large Language Model (LLM) Recommender

The prepared prompt is then sent to the Large Language Model (LLM). The prompt is crafted to include all relevant context about the user's question, such as any specific details, background information, and formatting instructions as retrieved by the retriever segment 3.1. The prompt also contains the user question and the behaviors defined in the prompt template. This comprehensive input allows the LLM to effectively analyze and synthesize a response that is relevant to the user's query and contextually appropriate. Once generated, the response is presented in plaintext format and displayed to the user through the frontend user interface, designed to ensure optimal readability and accessibility. This interaction is seamless and user-friendly, as the system is programmed to handle each inquiry with precision. Steps 3.A.1, 3.B.1, and 3.B.2 are executed every time a user poses a question, ensuring that each interaction adheres to a consistent protocol. This process involves capturing the user's input, processing it with the LLM, and delivering a response that meets the user's needs. The repetitive execution of these steps guarantees that the system remains responsive and efficient, continuously adapting to new questions and providing timely, accurate answers. This cycle facilitates efficient information retrieval and enhances the overall user experience by ensuring that responses are relevant and contextually aligned with the user's expectations.

IV. EXPERIMENTAL RESULTS

This section displays the experimental results by modifying different factors in the RAG pipeline. The test set consists of the questions and ground truth information shown in Figure. 3.

```
questions = [  
    "How many graduation requirements for graduate high school",  
    "How many weeks do I need to wait before I can switch classes?",  
    "What is the difference between core classes and electives",  
    "What pathway or courses should I take to become a computer scientist?",  
    "How do I enroll in a dual credit class?"  
]  
  
ground_truths = [  
    "26 credits (Foundation High School Program with Endorsement)",  
    "Five End of Course Exams in English I, English II, Algebra I, Biology and U.S. History",  
    ""Students must complete one of the following: submit a FAFSA application or submit a TASFA application or submit a signed opt-out form in Skyward in the parent portal"",  
    ""Students must also declare an endorsement.  
    They will automatically assigned the multidisciplinary endorsement if they don't declare an endorsement"",  
    "  
    "You can switch classes in the first week of school",  
    "You can level down from AP or Honors at the end of the sixth week",  
    "  
    "Core classes are mandatory courses",  
    ""Students must take four credits in Mathematics, including Algebra 2,  
    and four credits in science to graduate with a Distinguished Level of Achievement"",  
    "Elective classes are chosen by the student and certain number of courses are required to complete each endorsement",  
    "  
    ""Consider declaring as a STEM endorsement which requires:  
    4 credits of mathematics (Algebra II required)  
    4 credits of science (Chemistry and Physics required) and one of the following:  
    A coherent sequence of courses totaling 4 or more credits in Career and Technical Education (CTE) that includes at least 2 courses in the same career cluster and at least 1 advanced CTE course in a STEM qualifying career cluster/program of study.  
    Coherent sequence of four courses in Computer Science  
    Five credits in Mathematics including Algebra I, Geometry, Algebra II, and two additional courses which require Algebra II as a prerequisite  
    Five credits in Science including Biology, Chemistry, Physics, and two additional advanced Science courses  
    Complete a STEM qualifying Program of Study: At least three courses for four or more credits, and at least one advanced CTE course in a single program of study"",  
    "Consider taking advanced courses such as AP Computer Science Principles or AP Computer Science A",  
    "Consider taking advanced Math courses such as Algebra 2 and AP Calculus AB or AP Calculus BC"  
    ]
```

```
[  
    "Students in dual credit classes receive both college and highschool credit simultaneously",  
    "Students can choose to enroll for dual credit classes beginning the summer of sophomore year",  
    ""Classes not listed in the Prosper ISD course guide, but may be listed in the Collin College course catalog,  
    will not be approved for Dual Credit purposes"",  
    "Students are responsible for paying all tuition, book and fees for Dual Credit courses",  
    "Students must complete the application and registration process, including taking the TSIA."  
    ]
```

Fig. 3: Questions and Corresponding Ground Truths

Baseline Results - temperature = 0 and model = gpt-3.5-turbo

These results (Table 1) set the baseline results for the four metrics tested using the RAGAS framework: context precision, recall, faithfulness, and answer relevancy. These values will be compared with the below experiments.

Context precision measures how accurately the retrieved context aligns with the question. It evaluates the proportion of the retrieved context relevant to answering the question, focusing on the quality and precision of the information provided. Context Recall assesses the completeness of the retrieved context in addressing the question. It looks at the proportion of relevant information in the retrieved context compared to all possible relevant information, ensuring that no important details are omitted. Faithfulness evaluates how accurately the generated answer reflects the information present in the retrieved context. It checks whether the answer is true to the source material, avoiding any distortions or misinterpretations. Answer Relevancy measures the degree to which the generated answer is pertinent and directly related to the question. It focuses on ensuring that the answer is accurate and relevant to the specific inquiry.

Table 1: Baseline Results

Context Precision	Context Recall	Faithfulness	Answer Relevancy
1	0.8	1	0.8649
1	1	1	0.8094
0	0	0.8571	0.9639
1	0	0.5	0.9563
0.8056	1	1	0.9729

Experiment 1: temperature = 0.5

In this experiment, the model temperature was changed from 0 to 0.5. In the context of large language models (LLMs), temperature is a hyperparameter that controls the randomness of the model's output. It is used to adjust the probability distribution of the next word in the sequence, influencing the diversity and creativity of the generated text (Table 2).

Table 2: Modified Temperature Results

<i>Context Precision</i>	<i>Context Recall</i>	<i>Faithfulness</i>	<i>Answer Relevancy</i>
1	0.8	1	0.883
1	1	1	0.8094
0.3333	0	0.4286	0.975
1	0.3333	0.8571	0.9647
0.8056	1	0.875	0.9743

Experiment 2: *model_name= gpt-4o*

In this experiment, the base model that generated the response was changed from the gpt-3-turbo to the newer gpt-4o model. The gpt-3-turbo model is the faster of the two and is optimized for production use cases. The gpt-4o model is trained on a larger dataset and is claimed to be better at comprehension and reasoning tasks (Table 3), providing more relevant answers.

Table 3: Modified Model Results

<i>Context Precision</i>	<i>Context Recall</i>	<i>Faithfulness</i>	<i>Answer Relevancy</i>
1	0.75	1	0.8762
1	1	0.875	0.8094
0.75	0.3333	0.375	0.9894
1	0.333	0.9474	0.9659
1	1	1	0.9743

Discussion

The answer relevancy metric is an essential indicator in how useful the generated response answer would be to the user since relevancy is the key differentiator of a LLM chatbot system compared to traditional search mechanisms. The higher temperature experiment yielded higher answer relevancy scores than the baseline results for all questions in the dataset. The gpt-4o model experiment also yielded higher answer relevancy scores than the baseline results. This shows that the higher the creativity of the response (i.e. temperature or as an effect of the larger training data behind the gpt model), it will be more relevant to the user question.

The faithfulness metric is vital in ensuring that the answers are consistent with the retrieved context. In other words, this metric measures the system's tendency to hallucinate. As expected, the lower temperature results (baseline) performed better in this metric since restricting the sprawl of the LLM output ensures that the response is more closely synthesized from the retrieved context chunks. Figure 4 displays the averages of these two metrics across the different experiments, demonstrating the findings above.

Changing the models did not affect the faithfulness of the responses. The context recall and precision didn't change much between the experiments except for the model change from gpt-3-turbo to gpt-4o. Since the evaluation framework primarily outputs values of 1 or 0, Figure 5 compares the ratios of successfully meeting the context metrics (~1) or failing to meet the metrics (~0), with all values being rounded. The lack of variance in the bars across the experiments supports the above statements.

Note that these tests currently measure our model against itself with adjusted parameters. This approach was taken to identify the optimal output settings based on our current retrieval and generation chains. However, as this project is still a work in progress, we plan to benchmark our model against existing systems and gather student feedback for our next iteration. Moving forward, we can improve the precision and performance of context retrieval by incorporating a more diverse set of knowledge sources, such as student profile documents, alongside the course guide. Expanding our naive RAG architecture and experimenting with different LLM models like Mistral will enhance the responses' quality and accuracy.

Average of Generation Metrics

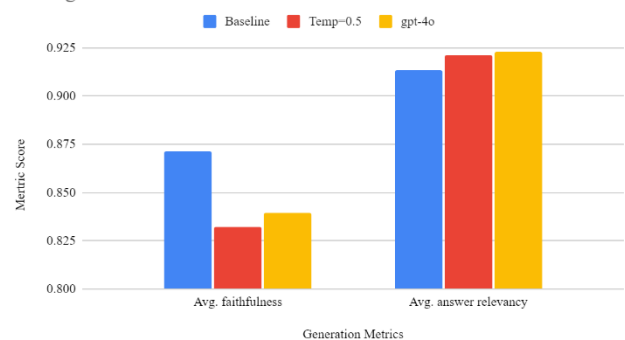


Fig. 4. Comparing Average of Generation Metrics

Ratio context recall and Ratio context precision

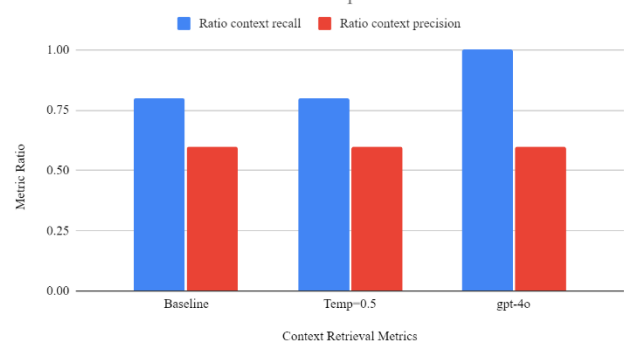


Fig. 5. Comparing Ratio of Retrieval Metrics

V. CONCLUSION

This research presents a novel AI-powered chatbot designed to provide personalized guidance to high school students. By leveraging a Large Language Model (LLM), the chatbot effectively synthesizes information from course guides and offers recommendations to help students navigate their educational journey. The chatbot's ability to

provide personalized guidance, enhance decision-making, and improve understanding makes it a valuable tool for students. Future enhancements, such as real-time data integration, login and persistent memory, granular control over the retrieval chain, and locally hosted LLM, can further improve the chatbot's capabilities and make it an even more indispensable resource for students. Overall, the chatbot has the potential to transform how students navigate their high school experience by providing reliable, personalized, and continuously improving support..

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